

SwiftParcel — Business Analysis Project

End-to-End Data Analysis Project

Google Sheets • SQL • Power BI

1. Project Overview

SwiftParcel is a synthetic logistics dataset created to analyse order performance, customer behaviour, delivery operations and revenue trends.

The project contains 60,000 orders, 12,000 customers and 10 warehouses covering January 2025 to August 2026.

The objective was to transform raw operational data into meaningful business insights and recommendations using Google Sheets, SQL and Power BI.

The analysis focused on key business questions including:

- How is revenue performing over time?
- Which customer groups and regions generate the most revenue?
- Which delivery methods and warehouses perform best?
- How significant are delivery delays and customer complaints?
- What patterns can be identified to support operational and commercial decision-making?
- What is the expected revenue trend for the following months?

The project follows an end-to-end Business Analyst workflow: validating and analysing the data, using SQL to investigate business questions, creating calculations and forecasts in Google Sheets, and presenting key findings through an interactive Power BI dashboard.

2. Dataset & Data Structure

Dataset

The SwiftParcel dataset contains:

- 60,000 orders
- 12,000 customers
- 10 warehouses
- Order, customer, warehouse and delivery information
- Data covering January 2025 to August 2026

The main order-level data includes order dates, delivery methods, product categories, order values, delivery costs, expected and actual delivery performance, customer ratings, complaints and order status.

Data Structure

The project uses a simple star-schema approach to organise the data.

Fact table

- Orders_Raw — contains individual order transactions and measurable business metrics such as order value, delivery cost and delay information.

Dimension tables

- Customers — contains customer attributes such as customer type, age group and region.
- Warehouses — contains warehouse attributes such as warehouse name, region, capacity and location type.

A star schema separates transactional data from descriptive information, making the dataset easier to analyse and reducing unnecessary duplication.

The main relationships are based on customer ID and warehouse ID. This structure was used consistently across the SQL and Power BI analysis.

3. Data Preparation & Validation

Before carrying out the analysis, the dataset was reviewed and validated to ensure that the data could be used reliably for business reporting.

The validation process included:

- Checking the structure and fields of the dataset.
- Confirming the number of records in the main tables.
- Reviewing data types and column names.
- Checking for potential data-quality issues.
- Validating key totals and calculations against the source data.
- Checking relationships between orders, customers and warehouses.
- Cross-checking key results between Google Sheets, SQL and Power BI.

The main Orders_Raw table contains 60,000 order records. The Customers table contains 12,000 customers and the Warehouses table contains 10 warehouses.

Key business totals were cross-checked across the different analysis tools. For example, total revenue was consistently calculated as £4,077,384.08 and total orders as 60,000.

This validation helped ensure that the subsequent analysis and dashboard outputs were based on consistent and reliable calculations.

4. Excel / Google Sheets Analysis

Analysis Approach

Excel / Google Sheets was used for exploratory data analysis, KPI calculations and revenue forecasting. For this project, the analysis was completed using Google Sheets.

The analysis included:

- Total orders and total revenue
- Average order value
- Customer type analysis
- Delivery method performance
- Product category performance
- Regional and warehouse performance
- Delivery delays and complaint rates
- Customer satisfaction
- Top customers by revenue and order frequency
- Monthly revenue trends
- Revenue forecasting

Key Findings

The dataset generated £4.08 million in revenue across 60,000 orders, with an average order value of £67.96.

Returning customers accounted for 61.35% of orders and generated £2.50 million in revenue, compared with 38.65% of orders and £1.58 million from new customers.

Delivery performance was an important area identified through the analysis. 44.10% of orders were delayed, while the overall customer complaint rate was 3.42%.

Monthly revenue remained relatively stable, generally around £200K per month. A trend-based forecast estimated revenue of approximately £206K per month for September to December 2026.

Forecasting

A monthly revenue forecast was created using historical revenue trends.

The forecast estimated:

- September 2026 — £206,030
- October 2026 — £206,236

- November 2026 — £206,442
- December 2026 — £206,648

The forecast suggests relatively stable revenue with a modest upward trend.

5. SQL Analysis

Analysis Approach

SQL was used to query the order, customer and warehouse data and answer specific business questions. The analysis was performed using SQLite.

The SQL analysis included aggregation, filtering, grouping, sorting, calculated metrics and joins between related tables.

A total of 26 SQL queries were developed to investigate business performance across revenue, customers, delivery operations, products and warehouses.

The analysis was designed to move beyond simple data retrieval and translate query results into business insights.

5.1 Executive KPI Summary

Business Question

What are the overall key performance indicators for SwiftParcel?

SQL Query

```
SELECT
    COUNT(*) AS total_orders,
    ROUND(SUM(order_value), 2) AS total_revenue,
    ROUND(AVG(order_value), 2) AS average_order_value,
    SUM(delayed_flag) AS delayed_orders,
    ROUND(100.0 * SUM(delayed_flag) / COUNT(*), 2) AS delay_rate_percent,
    ROUND(AVG(customer_rating), 2) AS average_customer_rating,
    SUM(complaint_flag) AS total_complaints,
    ROUND(100.0 * SUM(complaint_flag) / COUNT(*), 2) AS complaint_rate_percent
FROM Orders_Raw;
```

Result

KPI	Result
Total Orders	60,000
Total Revenue	£4,077,384.08

Average Order Value	£67.96
Delayed Orders	26,462
Delay Rate	44.10%
Average Customer Rating	4.08 / 5
Total Complaints	2,051
Complaint Rate	3.42%

What the Query Does

The query uses aggregate functions such as **COUNT**, **SUM** and **AVG** to calculate the main business KPIs from the Orders_Raw table. It also calculates the percentage of orders that were delayed and the percentage that resulted in a complaint.

Business Insight

SwiftParcel generated £4.08 million in revenue from 60,000 orders, with an average order value of £67.96. While customer satisfaction remained relatively strong at 4.08 out of 5, delivery performance was an important area for improvement, with 44.10% of orders recorded as delayed.

The results suggest that operational efficiency, particularly delivery performance, should be monitored alongside revenue and customer satisfaction.

5.2 Regional Performance

Business Question

Which regions generate the most orders and revenue?

SQL Query

```
SELECT
    region,
    COUNT(*) AS total_orders,
    SUM(order_value) AS total_revenue,
    AVG(order_value) AS average_order_value
FROM Orders_Raw
GROUP BY region
ORDER BY total_revenue DESC;
```

Result

Region	Orders	Revenue	Average Order Value
London	14,696	£994,582.38	£67.68
South East	8,562	£582,501.10	£68.03
North West	7,749	£527,264.35	£68.04
West Midlands	7,288	£495,787.45	£68.03
Yorkshire	6,346	£436,067.77	£68.72
East of England	6,005	£408,371.33	£68.01
South West	5,270	£356,860.83	£67.72
Scotland	4,084	£275,948.87	£67.57

What the Query Does

The query groups orders by region and calculates the number of orders, total revenue and average order value for each region. The results are then sorted from highest to lowest revenue.

Business Insight

London was the strongest region, generating approximately £995K in revenue from 14,696 orders. The South East and North West followed based on revenue.

Average order values were relatively consistent across all regions, suggesting that the differences in total revenue are primarily driven by order volume rather than significantly different customer spending levels.

5.3 Customer Type Performance

Business Question

How do new and returning customers compare in terms of orders, revenue and average order value?

SQL Query

```
SELECT
    c.customer_type,
    COUNT(o.customer_id) AS total_orders,
    SUM(o.order_value) AS total_revenue,
    AVG(o.order_value) AS average_order_value
```

```

FROM Orders_Raw o
JOIN Customers c
  ON o.customer_id = c.customer_id
GROUP BY c.customer_type
ORDER BY total_revenue DESC;

```

Result

Customer Type	Orders	Revenue	Average Order Value
Returning	36,897	£2,501,475.41	£67.80
New	23,103	£1,575,908.67	£68.21

What the Query Does

The query joins the Orders_Raw and Customers tables using the customer ID, then groups the results by customer type. It calculates total orders, total revenue and average order value for new and returning customers.

Business Insight

Returning customers generated the majority of SwiftParcel's activity, accounting for 61.35% of orders and approximately £2.50 million in revenue. New customers accounted for 38.65% of orders and approximately £1.58 million in revenue.

Average order values were very similar between the two groups, suggesting that the main difference in revenue is driven by the higher number of orders from returning customers rather than significantly higher spending per order.

5.4 Warehouse Delivery Performance

Business Question

Which warehouses have the highest delivery delay rates?

SQL Query

```

SELECT
  w.warehouse_name,
  w.region,
  COUNT(o.order_id) AS total_orders,
  SUM(o.delayed_flag) AS delayed_orders,
  ROUND(100.0 * SUM(o.delayed_flag) / COUNT(*), 2) AS delay_rate_percent,
  ROUND(AVG(o.delay_days), 2) AS average_delay_days
FROM Orders_Raw o
JOIN Warehouses w
  ON o.warehouse_id = w.warehouse_id

```

```
GROUP BY
  w.warehouse_name,
  w.region
ORDER BY delay_rate_percent DESC;
```

Result

Warehouse	Region	Orders	Delayed Orders	Delay Rate	Avg. Delay
London North	London	7,396	4,041	54.64%	0.98 days
Birmingham Central	West Midlands	7,288	3,369	46.23%	0.73 days
Leeds	Yorkshire	6,346	2,719	42.85%	0.64 days
Brighton	South East	8,562	3,661	42.76%	0.65 days
Liverpool	North West	3,814	1,613	42.29%	0.63 days
London South	London	7,300	3,054	41.84%	0.62 days
Manchester	North West	3,935	1,641	41.70%	0.62 days
Glasgow	Scotland	4,084	1,699	41.60%	0.62 days
Bristol	South West	5,270	2,192	41.59%	0.62 days
Cambridge	East of England	6,005	2,473	41.18%	0.63 days

What the Query Does

The query joins the Orders_Raw table with the Warehouses table to identify delivery performance by warehouse. It calculates the number of orders, delayed orders, delay rate and average delay for each warehouse.

Business Insight

London North is a clear operational outlier, with a delay rate of **54.64%**, compared with approximately 41–46% across the other warehouses.

This suggests that London North should be prioritised for further operational investigation. Potential areas to review include warehouse capacity, staffing, order volumes, processing times and local delivery operations.

5.5 Delivery Method Performance

Business Question

Which delivery methods are used most frequently, and how do they compare in revenue, average order value and delivery delays?

SQL Query

```
SELECT
    delivery_method,
    COUNT(*) AS total_orders,
    SUM(order_value) AS total_revenue,
    ROUND(AVG(order_value), 2) AS average_order_value,
    ROUND(100.0 * SUM(delayed_flag) / COUNT(*), 2) AS delay_rate_percent
FROM Orders_Raw
GROUP BY delivery_method
ORDER BY total_orders DESC;
```

Result

Delivery Method	Orders	Revenue	Average Order Value	Delay Rate
Home Delivery	28,673	£1,938,157.61	£67.60	44.46%
Locker	18,050	£1,226,415.71	£67.95	43.55%
Collection Point	7,217	£499,603.19	£69.23	44.05%
Express	6,060	£413,207.57	£68.19	44.13%

What the Query Does

The query groups orders by delivery method and calculates order volume, total revenue, average order value and delivery delay rate for each method.

Business Insight

Home Delivery was the most widely used option, accounting for almost half of all orders and generating approximately £1.94 million in revenue.

However, delivery delay rates were very similar across all four methods, ranging from **43.55% to 44.46%**. This suggests that delays may be driven by broader operational factors rather than being specific to one delivery method.

Collection Point had the highest average order value at £69.23, although the difference between delivery methods was relatively small.

5.6 Monthly Revenue Performance

Business Question

How has revenue performed over time, and are there any significant monthly trends?

SQL Query

```
SELECT
    strftime('%Y-%m', order_date) AS month,
    COUNT(*) AS total_orders,
    ROUND(SUM(order_value), 2) AS total_revenue,
    ROUND(AVG(order_value), 2) AS average_order_value
FROM Orders_Raw
GROUP BY strftime('%Y-%m', order_date)
ORDER BY month;
```

Result

Period	Revenue
Jan 2025	£204,958
Feb 2025	£187,638
Mar 2025	£203,955
Apr 2025	£206,489
May 2025	£203,507
Jun 2025	£206,630
Jul 2025	£206,577
Aug 2025	£211,451
Sep 2025	£208,080
Oct 2025	£203,490
Nov 2025	£193,444
Dec 2025	£205,019
Jan 2026	£209,838
Feb 2026	£186,721
Mar 2026	£214,906
Apr 2026	£198,465
May 2026	£205,616

Jun 2026 £199,917

Jul 2026 £207,996

Aug 2026 £212,687

What the Query Does

The query converts the order date into a year-month format and groups the orders by month. It calculates monthly order volume, total revenue and average order value to identify changes in performance over time.

Business Insight

Monthly revenue remained relatively stable throughout the period, with average monthly revenue of approximately **£203.9K**.

Revenue generally remained around £200K per month, although there were some fluctuations. The highest monthly revenue was approximately **£214.9K in March 2026**, while the lowest was approximately **£186.7K in February 2026**.

Overall, the results indicate a relatively stable revenue base rather than a strong upward or downward trend. This provides a useful foundation for the revenue forecasting carried out in Google Sheets.

6. Power BI Dashboard

Dashboard Purpose

Power BI was used to transform the analysis into an interactive business dashboard that provides a clear overview of SwiftParcel's performance.

The dashboard was designed to help users quickly understand key performance indicators, revenue trends, customer behaviour and delivery performance.

The dashboard includes:

- Total orders
- Total revenue
- Average order value
- Delivery performance
- Customer and regional analysis
- Revenue trends
- Interactive year filtering

The dashboard was structured across two pages to keep the information clear and easy to interpret.

6.1 Dashboard Overview

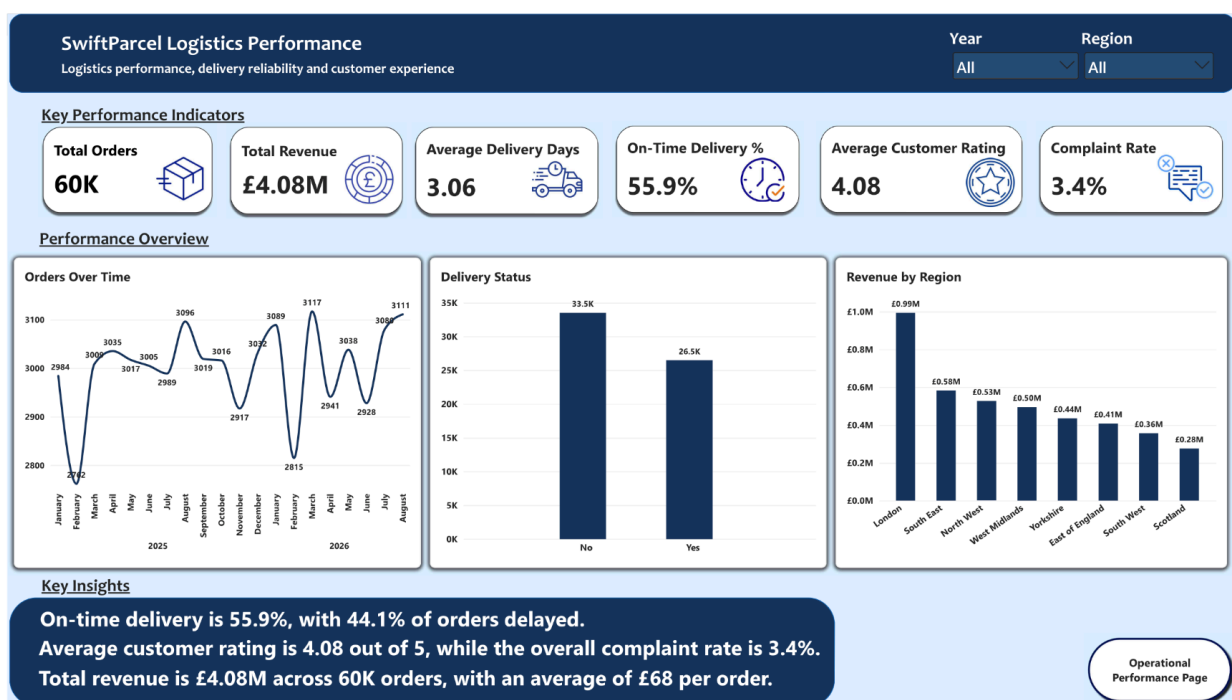
The Power BI dashboard brings the analysis together into an interactive business reporting tool.

Rather than presenting individual calculations separately, the dashboard allows a user to quickly assess overall performance and explore areas such as revenue, customer behaviour and delivery performance.

Interactive filters allow users to change the view of the data and compare performance across different periods.

The dashboard was designed with a focus on **clarity, usability and decision-making**, allowing business users to identify trends and potential areas for improvement without needing to work directly with the underlying data.

Power BI Dashboard





SwiftParcel Power BI dashboard — interactive view of business performance and operational insights.

7. Key Business Insights

The analysis across Google Sheets, SQL and Power BI identified several key findings relating to revenue, customers and operational performance.

7.1 Revenue Performance

SwiftParcel generated **£4.08 million in revenue across 60,000 orders**, with an average order value of approximately **£67.96**.

Monthly revenue remained relatively stable throughout the period, generally around £200K per month. This indicates a consistent revenue base, although there were some month-to-month fluctuations.

7.2 Customer Performance

Returning customers generated approximately **£2.50 million in revenue**, accounting for 61.35% of total orders.

New customers generated approximately **£1.58 million**, representing 38.65% of orders.

Average order values were very similar between the two groups, suggesting that the higher revenue from returning customers is mainly driven by their greater order frequency.

7.3 Delivery Performance

Delivery performance was identified as the main operational area requiring attention.

Overall, **44.10% of orders were delayed**. While delay rates were relatively similar across delivery methods, there was a significant difference between individual warehouses.

London North had the highest delay rate at 54.64%, making it a clear operational outlier compared with the other warehouses.

7.4 Customer Experience

The overall average customer rating was **4.08 out of 5**, while the complaint rate was **3.42%**.

This suggests that overall customer satisfaction was relatively positive, but continued monitoring of delivery performance is important because delays can potentially affect customer experience.

7.5 Key Takeaway

The analysis indicates that SwiftParcel has a stable revenue base and a strong contribution from returning customers. However, delivery delays represent the clearest operational improvement opportunity, particularly at the London North warehouse.

8. Recommendations

Based on the analysis, the following recommendations could support SwiftParcel's operational and commercial performance.

8.1 Investigate London North Warehouse Delays

London North had a significantly higher delay rate of **54.64%**, making it the clearest operational priority.

SwiftParcel should investigate potential causes such as warehouse capacity, staffing levels, order processing times and local delivery constraints. Performance should then be monitored against other warehouses to assess whether improvements are achieved.

8.2 Monitor Delivery Performance

With **44.10% of orders delayed overall**, delivery performance should be tracked as a key operational KPI.

Management could monitor delay rates regularly by warehouse, region and delivery method to identify emerging issues and measure the impact of operational improvements.

8.3 Strengthen Customer Retention

Returning customers generated **61.35% of orders and £2.50 million in revenue**.

SwiftParcel could continue to focus on customer retention through loyalty initiatives, personalised offers and reliable service, while also using insights from existing customers to support new-customer conversion.

8.4 Use Revenue Forecasting for Planning

Revenue remained relatively stable, with the forecast suggesting approximately **£206K per month** for September to December 2026.

This type of forecasting could support resource planning, budgeting and operational decision-making, while actual performance should continue to be compared with forecast results.

8.5 Continue Data-Driven Monitoring

The analysis demonstrates the value of combining operational, customer and financial data.

SwiftParcel could continue using dashboards and regular reporting to monitor KPIs, identify performance changes and support evidence-based business decisions.

9. Tools & Skills

Tools Used

- **Google Sheets / Excel** — data analysis, KPI calculations, pivot analysis and revenue forecasting
- **SQL (SQLite)** — data querying, aggregation, joins, calculated metrics and business analysis
- **Power BI** — interactive dashboards, KPI reporting and data visualisation

Business Analysis Skills Demonstrated

- Data validation and quality checking
- Exploratory data analysis
- KPI development and performance analysis
- Customer and operational analysis
- Revenue trend analysis and forecasting
- SQL querying and data investigation
- Data visualisation and dashboard development
- Identifying business problems and opportunities
- Translating data into actionable business insights
- Making evidence-based recommendations